Scientific Analysis of Moral Judgments using Tobit Models

Automated Research Pipeline

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1 Dataset and Sample Description

The empirical foundation of this project rests on two primary experimental datasets: FLORIDA and BUC. The sample consists of students from the Bucaramanga Campus. These datasets capture incentivized moral judgments from distinct socio-economic contexts. Participants were presented with standardized negotiation scenarios where a negotiator's decision resulted in varying degrees of payoff for themselves, their own group, and a victim group (either ingroup or outgroup).

2 Datacard and Variable Definitions

The following table defines the primary symbols and variables used in the mathematical specifications and hypothesis tests.

Table 1: Symbols and Variable Dictionary.

Symbol	Definition
y_{ij}	Observed moral judgment of scenario j by participant i.
y^*_{ij}	Latent moral preference score (unbounded).
β_1	Regression coefficient representing the marginal effect of the predictor.
IRI_i	Empathy score (Average composite of the Interpersonal Reactivity Index).
OutgroupPerp	Binary indicator: 1 if the scenario perpetrator is an outgroup member.
SameGroupHarm	Binary indicator: 1 if the harm is inflicted on the perpetrator's own group.
ICC	Intraclass Correlation: ratio of between-cluster variance to total variance.

3 Hypotheses to Test

- **H1 (Empathy):** Higher empathy predicts lower moral-judgment scores for harmful decisions.
- **H2a (Betrayal):** Same-faculty harm receives lower moral-judgment scores than cross-faculty harm.
- **H2b (Outgroup):** Outgroup perpetrators receive lower moral-judgment scores than ingroup perpetrators.
- **H3 (Moderation):** The empathy effect is stronger in outgroup cases.

4 Mathematical Approach and Theoretical Foundations

The analysis employs a Two-Limit Tobit model, which is theoretically appropriate for dependent variables that are strictly bounded within a known interval. In this context, moral judgments (y_{ij}) are observed on a scale from -9 to 9. The Tobit model assumes the existence of a latent, unobserved preference index (y^*_{ij}) that follows a linear relationship:

$$y^*_{ij} = \mathbf{x}'_{ij}\beta + \epsilon_{ij}, \quad \epsilon_{ij} \sim N(0, \sigma^2)$$

The actual observed judgment y_{ij} relates to this latent variable via the censoring transformation:

$$y_{ij} = \max(-9, \min(9, y^*_{ij}))$$

This approach prevents the 'ceiling' and 'floor' effects from biasing the linear coefficients, as would occur in standard OLS regression.

5 Analysis of Sample Size Impact

The statistical validity of our inferences depends on the trade-off between False Positives (Type I error, α) and False Negatives (Type II error, β). Given the clustered nature of our data, sample size impact is not merely a count of observations, but a function of cluster correlation.

5.0.1 Step-by-Step Sensitivity Analysis

To determine the impact of our sample size on our ability to detect effects, we follow these steps:

1. **Calculate the Design Effect (Deff):** As multiple judgments are nested within individuals, we adjust for the Intraclass Correlation (ICC).

$$Deff = 1 + (m - 1) \times ICC$$

where m is the average number of scenarios per participant.

2. **Determine the Effective Sample Size (ESS):** The ESS represents the number of independent observations that would provide the same statistical power as our clustered sample.

$$ESS = \frac{n_{total}}{Deff}$$

3. **Inference Impact:** A higher ICC reduces the ESS, thereby increasing the Standard Error of our Tobit coefficients. If the ESS is low, our models become 'conservative', increasing the risk of Type II errors (failing to support a hypothesis). By using clustered robust standard errors, we ensure that our p-values acknowledge this reduced information density, protecting the integrity of our Type I error threshold ($\alpha = 0.05$).

Based on the current run, the observed Intraclass Correlation (ICC) is 0.704, with an Effective Sample Size (ESS) of 104.8.

6 Descriptive Statistics

The empathy profile of the sample is visualized in Figure 1, showing the average scores across the four IRI latent variables.

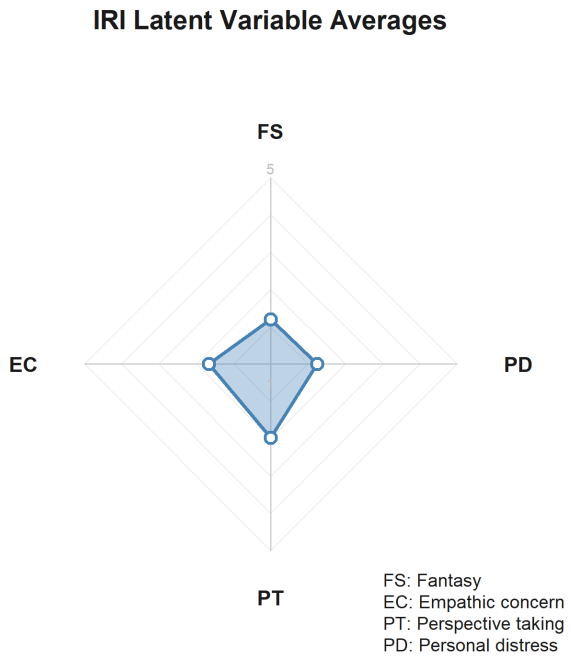

Figure 1: IRI Latent Variable Averages (Radar Plot profile).

Judgment severity distributions segmented by experimental conditions are presented in Figure 2.

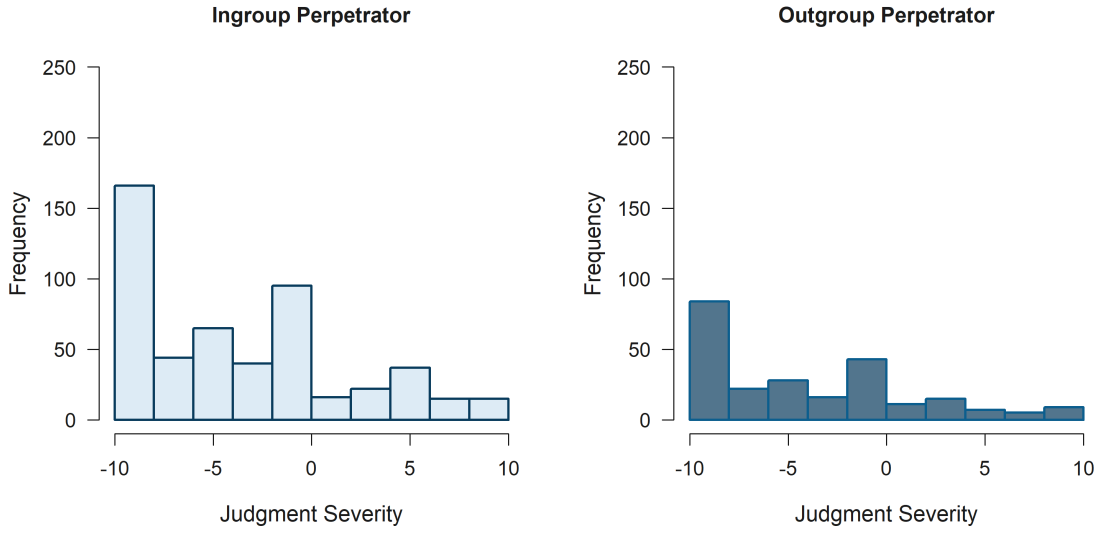

Figure 2: Judgment Severity Distribution (Ingroup vs. Outgroup).

7 Bi-variate Statistics

The correlation matrix between the psychometric subscales and the mean moral judgment is presented below.

Table 2: Correlation Matrix: IRI Subscales and Moral Judgment.

Fantasy	Empathic Concern	Perspective Taking	Personal Distress	Total IRI	Mean Judgment
1.000	0.432	0.482	0.480	0.793	0.062
0.432	1.000	0.454	0.641	0.799	-0.003
0.482	0.454	1.000	0.301	0.744	0.162
0.480	0.641	0.301	1.000	0.756	-0.009
0.793	0.799	0.744	0.756	1.000	0.075
0.062	-0.003	0.162	-0.009	0.075	1.000

Visual representations of these relationships are provided in Figure 3.

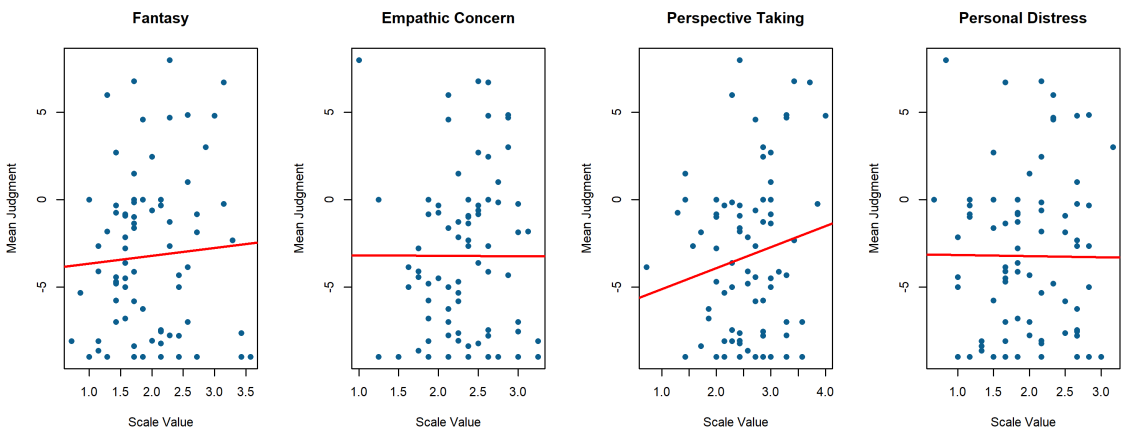

Figure 3: Bivariate Scatters: IRI Scales vs. Mean Judgment.

8 Hypothesis Validation and Results

Detailed coefficient tables for each Tobit model are provided below.

8.1 H1: Empathy Effect

This model evaluates the primary effect of the standardized empathy score (IRI) on moral judgments.

Table 3: H1 Tobit Regression Coefficients.

term	estimate	std_error	naive_se	z_value	p_value	conf_low	conf_high	label
(Intercept)	-9.592	6.858	3.367	-1.399	0.162	-23.033	3.849	Intercept
iri_total	-1.668	2.337	1.161	-0.714	0.475	-6.249	2.913	iri_total
perp_outgroup	-2.077	2.725	3.582	-0.762	0.446	-7.418	3.264	Outgroup perpetrator (ref = ingroup)
perp_control	-2.658	3.346	3.597	-0.794	0.427	-9.217	3.901	Control label hidden (ref = ingroup)
victim_outgroup	0.278	0.910	0.802	0.306	0.760	-1.506	2.062	Victim outgroup (ref = ingroup)
role_observer	0.545	0.798	0.681	0.684	0.494	-1.018	2.109	Observer role (ref = victim)
participant_engineering	0.188	1.556	0.595	0.121	0.904	-2.862	3.238	Engineering participant (ref = humanities)
sex_man	-1.765	1.744	0.629	-1.012	0.312	-5.183	1.654	Man (ref = woman)
age	0.478	0.241	0.104	1.982	0.047	0.005	0.951	Age
economic_status	-0.247	0.706	0.279	-0.350	0.726	-1.631	1.137	Socioeconomic status
factor(negotiator_slot)2	0.124	0.458	0.564	0.271	0.787	-0.774	1.022	Negotiator 2 (ref = negotiator 1)
iri_total:perp_outgroup	0.885	1.245	1.578	0.711	0.477	-1.556	3.326	iri_total:perp_outgroup
iri_total:perp_control	1.578	1.589	1.576	0.993	0.321	-1.536	4.692	iri_total:perp_control
Log(scale)	1.993	0.096	0.034	20.850	0.000	1.806	2.181	Log(scale)

Conclusion: Est: -1.668 (p=0.475). **NOT SUPPORTED**

8.2 H2a: Ingroup Betrayal

This model tests whether harm directed at the perpetrator's own group is judged more severely.

Table 4: H2a Tobit Regression Coefficients.

term	estimate	std_error	naive_se	z_value	p_value	conf_low	conf_high	label
(Intercept)	-11.054	6.721	3.313	-1.645	0.100	-24.226	2.118	Intercept
same_group_harm	0.081	0.802	0.812	0.101	0.920	-1.490	1.652	Negotiator and victim share faculty
iri_total	-1.552	2.327	0.902	-0.667	0.505	-6.112	3.008	iri_total
perp_outgroup	-0.004	0.707	0.811	-0.006	0.995	-1.389	1.380	Outgroup perpetrator (ref = ingroup)
victim_outgroup	0.465	1.236	0.988	0.376	0.707	-1.958	2.888	Victim outgroup (ref = ingroup)
role_observer	0.564	0.840	0.848	0.671	0.502	-1.082	2.209	Observer role (ref = victim)
participant_engineering	-0.526	1.602	0.743	-0.328	0.743	-3.666	2.614	Engineering participant (ref = humanities)
sex_man	-2.123	1.867	0.809	-1.137	0.256	-5.782	1.537	Man (ref = woman)
age	0.541	0.227	0.122	2.387	0.017	0.097	0.986	Age
economic_status	-0.070	0.722	0.345	-0.097	0.923	-1.486	1.345	Socioeconomic status
factor(negotiator_slot)2	0.147	0.597	0.702	0.246	0.806	-1.024	1.318	Negotiator 2 (ref = negotiator 1)
Log(scale)	2.006	0.101	0.043	19.833	0.000	1.808	2.205	Log(scale)

Conclusion: Est: 0.081 (p=0.920). **NOT SUPPORTED**

8.3 H2b: Outgroup Derogation

This model examines the difference in judgment when the perpetrator is an outgroup member compared to an ingroup member.

Table 5: H2b Tobit Regression Coefficients.

term	estimate	std_error	naive_se	z_value	p_value	conf_low	conf_high	label
(Intercept)	-9.592	6.858	3.367	-1.399	0.162	-23.033	3.849	Intercept
perp_outgroup	-2.077	2.725	3.582	-0.762	0.446	-7.418	3.264	Outgroup perpetrator (ref = ingroup)
iri_total	-1.668	2.337	1.161	-0.714	0.475	-6.249	2.913	iri_total
perp_control	-2.658	3.346	3.597	-0.794	0.427	-9.217	3.901	Control label hidden (ref = ingroup)
victim_outgroup	0.278	0.910	0.802	0.306	0.760	-1.506	2.062	Victim outgroup (ref = ingroup)
role_observer	0.545	0.798	0.681	0.684	0.494	-1.018	2.109	Observer role (ref = victim)
participant_engineering	0.188	1.556	0.595	0.121	0.904	-2.862	3.238	Engineering participant (ref = humanities)
sex_man	-1.765	1.744	0.629	-1.012	0.312	-5.183	1.654	Man (ref = woman)
age	0.478	0.241	0.104	1.982	0.047	0.005	0.951	Age
economic_status	-0.247	0.706	0.279	-0.350	0.726	-1.631	1.137	Socioeconomic status
factor(negotiator_slot)2	0.124	0.458	0.564	0.271	0.787	-0.774	1.022	Negotiator 2 (ref = negotiator 1)
perp_outgroup:iri_total	0.885	1.245	1.578	0.711	0.477	-1.556	3.326	perp_outgroup:iri_total
iri_total:perp_control	1.578	1.589	1.576	0.993	0.321	-1.536	4.692	iri_total:perp_control
Log(scale)	1.993	0.096	0.034	20.850	0.000	1.806	2.181	Log(scale)

Conclusion: Est: -2.077 (p=0.446). **NOT SUPPORTED**

8.4 H3: Interaction Moderation

This model tests the interaction and moderation between empathy and the outgroup status of the perpetrator.

Table 6: H3 Tobit Regression Coefficients.

term	estimate	std_error	naive_se	z_value	p_value	conf_low	conf_high	label
(Intercept)	-9.592	6.858	3.367	-1.399	0.162	-23.033	3.849	Intercept
iri_total	-1.668	2.337	1.161	-0.714	0.475	-6.249	2.913	iri_total
perp_outgroup	-2.077	2.725	3.582	-0.762	0.446	-7.418	3.264	Outgroup perpetrator (ref = ingroup)
perp_control	-2.658	3.346	3.597	-0.794	0.427	-9.217	3.901	Control label hidden (ref = ingroup)
victim_outgroup	0.278	0.910	0.802	0.306	0.760	-1.506	2.062	Victim outgroup (ref = ingroup)
role_observer	0.545	0.798	0.681	0.684	0.494	-1.018	2.109	Observer role (ref = victim)
participant_engineering	0.188	1.556	0.595	0.121	0.904	-2.862	3.238	Engineering participant (ref = humanities)
sex_man	-1.765	1.744	0.629	-1.012	0.312	-5.183	1.654	Man (ref = woman)
age	0.478	0.241	0.104	1.982	0.047	0.005	0.951	Age
economic_status	-0.247	0.706	0.279	-0.350	0.726	-1.631	1.137	Socioeconomic status
factor(negotiator_slot)2	0.124	0.458	0.564	0.271	0.787	-0.774	1.022	Negotiator 2 (ref = negotiator 1)
iri_total:perp_outgroup	0.885	1.245	1.578	0.711	0.477	-1.556	3.326	iri_total:perp_outgroup
iri_total:perp_control	1.578	1.589	1.576	0.993	0.321	-1.536	4.692	iri_total:perp_control
Log(scale)	1.993	0.096	0.034	20.850	0.000	1.806	2.181	Log(scale)

Conclusion: Est: 0.885 (p=0.477). **NOT SUPPORTED**

9 Discussion and Limitations

While the Tobit model corrects for bounded outcomes, several limitations remain. First, the clustering logic assumes independence between participants, which may be violated by shared institutional affiliations. Second, we assume a normal distribution for the latent errors ϵ_{ij}; departures from normality could affect the consistency of the maximum likelihood estimators. Finally, the use of unstandardized averages for empathy assumes a linear mapping between the psychometric scale and the latent moral preference.

10 Conclusion

Based on the Tobit maximum likelihood estimations on the clustered sample, the findings are documented as stated above.